

The possible role of female sex hormones in milk from pregnant cows in the development of breast, ovarian and corpus uteri cancers.

The continued increase in incidence of some hormone-related cancers worldwide is of great concern. Although estrogen-like substances in the environment were blamed for this increase, the possible role of endogenous estrogens from food has not been widely discussed. We are particularly concerned about cows' milk, which contains a considerable quantity of estrogens. When we name cows' milk as one of the important routes of human exposure to estrogens, the general response of Western people is that "man has been drinking cows' milk for around 2000 years without apparent harm." However, the milk that we are now consuming is quite different from that consumed 100 years ago. Unlike their pasture-fed counterparts of 100 years ago, modern dairy cows are usually pregnant and continue to lactate during the latter half of pregnancy, when the concentration of estrogens in blood, and hence in milk, increases. The correlation of incidence and mortality rates with environmental variables in worldwide countries provides useful clues to the etiology of cancer. In this study, we correlated incidence rates for breast, ovarian, and corpus uteri cancers (1993-97 from Cancer Incidence in Five Continents) with food intake (1961-97 from FAOSTAT) in 40 countries. Meat was most closely correlated with the breast cancer incidence ($r=0.827$), followed by milk (0.817) and cheese (0.751). Stepwise multiple-regression analysis (SMRA) identified meat as the factor contributing most greatly to the incidence of breast cancer ($[R]=0.862$). Milk was most closely correlated with the incidence of ovarian cancer ($r=0.779$), followed by animal fats (0.717) and cheese (0.697). SMRA revealed that milk plus cheese make the greatest contribution to the incidence of ovarian cancer ($[R]=0.767$). Milk was most closely correlated with corpus uteri cancer ($r=0.814$), followed by cheese (0.787). SMRA revealed that milk plus cheese make the most significant contribution to the incidence of corpus uteri cancer ($[R]=0.861$). In conclusion, increased consumption of animal-derived food may have adverse effects on the development of hormone-dependent cancers. Among dietary risk factors, we are most concerned with milk and dairy products, because the milk we

drink today is produced from pregnant cows, in which estrogen and progesterone levels are markedly elevated.